

EXPERIENCE

Lawrence Livermore National Laboratory - Graduate Intern, Livermore, CA June 2025 - present

Faculty mentor: Jesse Norris - Nuclear Criticality Safety Division

- Developed CRISP (CRITICAL Simulation Pipeline) for parallelizing jobs from a radiation transport Monte Carlo code: COG
- Test driven Python development included Poetry, CI/CD pipeline, sqlite scalable materials database, MPI optimization, CLI development, feature branching, and Sphinx documentation
- Utilized world-class LLNL HPC clusters Dane and Ruby with Slurm sbatch/srun scripts
- CRISP successfully automated the materials compiling and k_{eff} benchmarking of ~ 3420 ICSBEP critical reactors

Brookhaven National Laboratory - Nuclear Science Intern, Upton, NY

2022 - 2024

Faculty mentor: Gustavo Nobre - National Nuclear Data Center

- Developed BRR (Bayesian Resonance Reclassifier) for heavy-nuclei resonance reclassification
- Applied python machine learning and Random Matrix Theory to develop synthetic training sets and classify resonance spin assignments for capture cross-sections
- Used the neutron transport code OpenMC to perform perturbative sensitivity analyses of the thermal $1/v$ neutron capture cross-sections for critical reactors

Truman State University - Student Astronomy Researcher, Kirksville, MO

2021 - 2022

Faculty mentor: Vayujeet Gokhale - Department of Physics

- Calculated trajectories of Starlink satellites to optimize telescope viewing plans
- Researched long-exposure luminosity data corruption due to Starlink satellite interference
- Developed a GUI for Truman astronomy students
- Wrote and submitted a proposal for the TruScholars grant